

Synthesis Report

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1 Problem Statement

Execute synthesis for 10 benchmarks listed as follows:

1. ISCAS-85: 1 design
2. CEP: 2 designs
3. ITC99: 2 designs
4. UART: 3 designs
5. CUSTOM: 2 designs

By using these 6 libraries as follows:

1. slow
2. fast
3. fast_hvt
4. slow_hvt
5. fast_lvt
6. slow_lvt

Using two frequency options:

1. 100MHz
2. 400MHz

With 50 % duty cycle each. Answer the following statements:

1. Comparison of area/timing/power for different library corners for each design
2. Comparison of area/timing/power as per each frequency
3. Identify the count of each cell type by reading the generated netlist
4. Identify the most power consuming sub-block within each benchmark
5. Identify the sub-block with highest area within each benchmark

Also include your approach to automating the execution and processing of tool reports/log files. Include any specific observations from the experiments, and your key learnings from this exercise.

2 Approach

We created a folder synthesis under which all the synthesis operation takes place. In order to make the process efficient , We have created multiple bash scripts as follows to automate the synthesis process.

2.1 Bash 1

```
#!/bin/bash
for ((i=1;i<11;i++))
do
    mkdir -p ~/synthesis/d$i/400Hz ~/synthesis/d$i/100Hz ~/synthesis/d$i/design
done
```

By invoking the above script, We create 10 folders named d1 - d10 which indicates the benchmarks inside the synthesis folders under which 3 folders named:

1. 100Hz
2. 400Hz
3. design

are created which respectively indicates the 100MHz option and the 400MHz option. The design folder contains the input hdl files.

2.2 Bash 2

```
S100=~/.100.sdc
S400=~/.400.sdc
for ((i=1;i<11;i++))
do
    cp $S100 ~/synthesis/d$i/100Hz/100Hz.sdc
    cp $S400 ~/synthesis/d$i/400Hz/400Hz.sdc
done
```

This bash script is used to copy the constraint files(.sdc files) into the repective 100Hz and 400Hz files according to the frequency used in the constraint file.(sdf files are later modified based on the design).

2.3 Bash 3

```
#!/bin/bash
libraries=("slow" "fast" "slow_lvt" "fast_lvt" "slow_hvt" "fast_hvt")
frequencies=("100Hz" "400Hz")

for dir in /DIG_DESIGN/INTERNS/dic_lab_12/synthesis/*
do
    rm -f "$temp_file"
    touch "$temp_file"
    temp_file="$dir/temp"
    design_dir="$dir/design"
    for file in "$design_dir"/*.v
    do
        echo -n "$(basename "$file") " >> "$temp_file"
    done
    for freq in "${frequencies[@]}"
    do
        for lib in "${libraries[@]}"
        do
            mkdir -p "$dir/$freq/$lib/output"
            mkdir -p "$dir/$freq/$lib/reports"
            lib_output_dir="$dir/$freq/$lib"
            tcl_file="$dir/$freq/synthesis_$lib.tcl"

            if [[ "$lib" == *"hvt"* ]]
            then
                lib_file="${lib/_hvt/}_vdd1v0_basicCells_hvt.lib"
            elif [[ "$lib" == *"lvt"* ]]
            then
                lib_file="${lib/_lvt/}_vdd1v0_basicCells_lvt.lib"
            else
                lib_file="${lib}_vdd1v0_basicCells.lib"
            fi
        done
    done
done
```

```

cat > "$tcl_file" << END
set_attribute init_lib_search_path /DIG_DESIGN/INTERNS/dic_lab_12/PDK_DIC
set_attribute init_hdl_search_path $dir/design
set_attribute library $lib_file

read_hdl $(cat "$temp_file")
elaborate
read_sdc ${dir}/${freq}/${freq}.sdc
set_attribute syn_generic_effort medium
set_attribute syn_map_effort medium
set_attribute syn_opt_effort medium

syn_generic
syn_map
#synthesize -to_mapped -effort medium
syn_opt

write_hdl -mapped > ${lib_output_dir}/output/counter_net.v
write_sdc > ${lib_output_dir}/output/counter_out.sdc

#Write results in Report folder
report_timing > ${lib_output_dir}/reports/report_timing.rpt
report_power > ${lib_output_dir}/reports/report_power.rpt
report_area > ${lib_output_dir}/reports/report_area.rpt
END
done
done
done

```

The above bash script is used to create .tcl files for all the libraries and frequencies(100MHz and 400MHz). Therefore, a total of $6 \times 2 \times 10 = 120$ (6 libraries, 2 frequencies and 10 designs) tcl files are written using the above script. It also creates 2 directories named output and reports for storing the extracted data after synthesis. (The above bash is only valid for .v files and other .vhd designs are done manually).

2.4 Bash 4

```
#!/bin/bash
```

```

for ((i=1;i<11;i++))
do
    benchmark=/DIG_DESIGN/INTERNS/dic_lab_12/synthesis/d${i}
    frequencies=("100Hz" "400Hz")
    for dir in "${frequencies[@]}"
    do
        dir_path="$benchmark"/"$dir"
        tcl_files=("synthesis_slow.tcl" "synthesis_fast.tcl" "synthesis_slow_lvts")
        for tcl_file in "${tcl_files[@]}"
        do

```

```
genus -legacy_ui -f "$dir_path"/"$tcl_file" -batch
```

```
done
done
done
```

The above script runs synthesis for all the benchmarks. It searches for 100MHz and 400MHz and inside that, it checks whichever file has 'tcl' extension in it and runs it.

3 Analysis

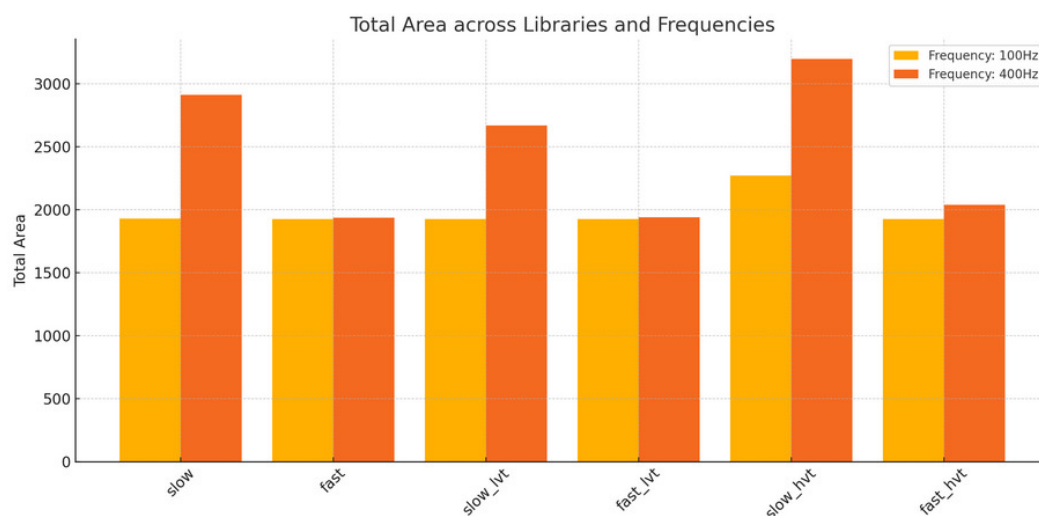


Figure 1: Total area for each library and frequency

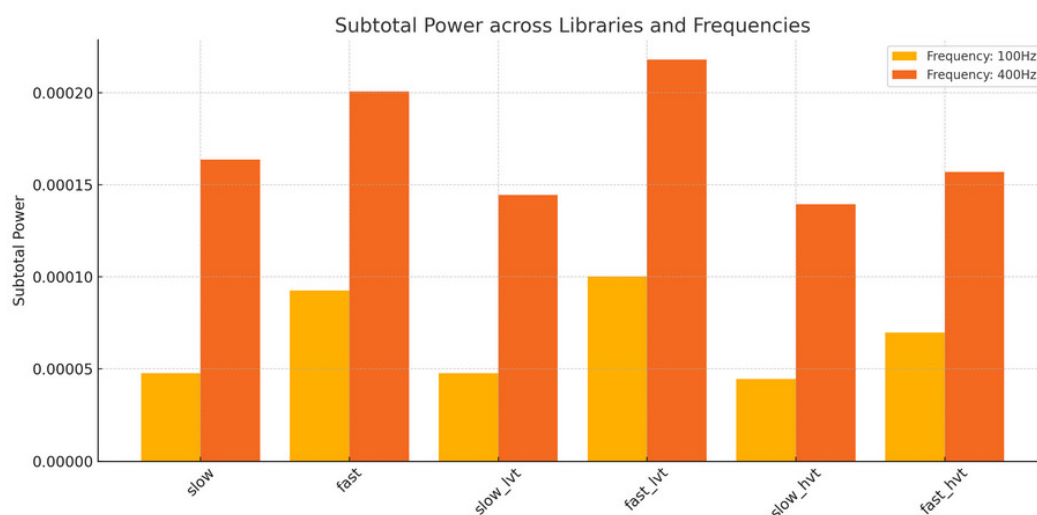


Figure 2: Total Power for each library and frequency

⇒ 400MHz consumes more power and area in general according to the above observations.